Nicholas Garza

nickgarza.vercel.app | nicholaspgarza@gmail.com | linkedin.com/in/nicholaspgarza

Experience

Microsoft

Software Engineer II - Xbox Graphics August 2020 - Present

- Designed and deployed a performance dashboard to capture FPS regressions on driver and chipset changes across handheld and desktop devices, used to demonstrate driver optimizations to leadership and external partners
- Automated the next generation Xbox infrastructure to host handheld and pre-release devices for driver build deployment and automated game benchmarking using Azure DevOps, Postgres SQL, Powershell, and Python
- Owned the Xbox DirectX runtime validation and implemented tooling to visualize execution timing in C++
- Enabled high frame rate support and HDR plane composition for the Xbox consoles in the kernel mode driver
- Instrumented the Xbox user mode driver to capture GPU command buffers and repeatedly submit work to track execution timing within nanoseconds across driver and shader compiler changes in C++
- Doubled mesh shader throughput by implementing a mixed-precision multiplication in GPU microcode
- Owned the GPU Pipeline View and implemented debug visualizations for alpha transparencies, color formatting, and texture resources for the GPU Performance Investigator for Xbox in HLSL, C++, and C#
- Mentored team interns and intern cohort career development workshops in Q&A sessions and morale events

Medtronic

Software Engineer Intern May 2019 - August 2019

- Developed an automated testing framework in Java for the Minimed insulin pump and integrated the design into the Azure based testing platform.
- Expanded the framework to showcase real time unit testing results and metadata visualization for Android/iOS mobile app crash reporting.

Johns Hopkins Computer Science Department

Head Teaching Assistant January 2019 - May 2020

- Implemented autograder support for grading student assignments in Python
- Lead team of course assistants to set learning goals for introductory programming course

University of Virginia

Systems Bioengineering Research Intern May 2018 - August 2018

- Analyzed terabytes of human genomic data in R to correlate inflammatory cell response with coronary artery diseases
- Developed workflows for filtering and sorting gene expression types in R and C++
- Accepted to present at 2018 BMES conference and 2018 REUS conference

Technical Skills

Programming Languages Python, C/C++, Java, C#, SQL, R

Tools/Frameworks Git, DirectX, *nix, Windbg, Matlab, JUnit

Education

Johns Hopkins University

Baltimore, MD

B.S. COMPUTER SCIENCE, B.S. BIOMEDICAL ENGINEERING August 2016 - May 2020

- Honors: CS Honors Graduate, Dean's List, Upsilon Pi Epsilon Honors Society, Hopkins Academic Success Award